ELVIA: First paper and report

Editable scientific template, version 2026-10-23. This is a template, not a completed review or an endorsement.

> Replace bracketed prompts with your own report. A focused check of one claim is sufficient; no minimum page count is required. Keep each problem numbered and use the headings below. Delete inapplicable optional sections. Compile with pdfLaTeX or XeLaTeX, or upload this single UTF-8 .tex document. Include all equations and references here; do not depend on external images, input files or bibliography files. If external material is essential to reading the report, submit a self-contained PDF instead.

Submission files

- This report: one PDF or one complete .tex file.
- Original article: one PDF.
- Main original supplementary document, if applicable: one separate PDF. Select Not applicable on the website when there is none.
- Optional verification code: one ZIP or 1–10 source-code/support files, at most 5 MB total, including unpacked ZIP contents. A short README with commands and dependencies is helpful; no prescribed folder structure is required.

The report, original article, original supplement and verification code have separate upload slots. Do not combine them into a report ZIP.

1 Paper identity and scope

Paper *[Exact title, authors, venue/year, DOI or official URL.]*

Version *[Published version, arXiv version number or precise dated version.]*

Materials inspected *[Main PDF version; supplement title/version, or none; public author-code repository and exact commit/version, if inspected.]*

Scope *[Which claims, sections, experiments or methods you checked and why.]*

Summary *[A short account of what your check establishes and what remains unresolved. Distinguish clarification, potential issue and demonstrated issue.]*

What was not checked *[Unexamined experiments, assumptions, datasets or claims.]*

2 Problem 1: *[short descriptive title]*

Repeat this entire section as Problem 2, Problem 3, etc. for additional independent issues. Do not split one issue into many fragments. A well-supported No issue found within scope result is also useful.

2.1 Location and assessment

Exact location *[PDF page and section; equation/figure/table number; or code file, function and lines plus repository commit.]*

Severity/outcome *[Choose exactly one: Minor; Major; Needs evidence; No issue found within scope.]*

- **Minor**: a local error or missing detail with bounded impact; do not claim the main conclusion fails without showing that consequence.
- **Major**: demonstrated effect on a central claim or essential result. Show the dependence between the issue and that claim.
- **Needs evidence**: a plausible concern whose validity or impact has not yet been established.
- **No issue found within scope**: the specified check supports the original argument. This is not a certificate for the whole paper.

2.2 Claim and assumptions

Original claim *[Brief exact quotation or faithful paraphrase, with location.]*

Conditions *[Definitions, assumptions, units, parameter range, sample or experimental unit, and which conditions are stated versus inferred.]*

2.3 Check and evidence

Method *[Step-by-step derivation, counterexample, calculation, experimental control, statistical check or code inspection. Explain why it tests the claim.]*

Result *[Actual observations or derived results. Separate analytic proof, finite numerical checks, empirical observations and conjecture.]*

For calculations, give the formula, symbols and substitutions, for example $r = x/x_0$, with units and the condition $x_0 \neq 0$. For data analyses, report sample counts, exclusions, independent units, preprocessing and uncertainty. For a computational check, give versions, commands, random seeds and tolerances where relevant, expected output and what you actually obtained. Do not imply a program was run when it was only inspected.

Evidence identifiers *[References, code filenames, output values or stable artifact links needed to inspect this check.]*

2.4 Impact, counterevidence and possible correction

Impact *[Exactly which result changes, under what conditions, and which claims remain unaffected. Justify the chosen severity.]*

Counterevidence *[The strongest alternative reading or contrary evidence you checked, and whether it resolves the concern.]*

Limits *[Unresolved steps, unavailable information, failed checks and what this evidence does not establish.]*

Possible correction (optional) *[A precise clarification, corrected statement or repair candidate. State what still needs verification; do not claim adoption.]*

3 Reproduction and overall conclusion

Verification materials *[List your optional uploaded code files or ZIP and what each does. If no code is needed, state that explicitly.]*

Reproduction *[Commands, dependencies and expected output, or Not applicable for a self-contained mathematical/textual check. Do not include credentials.]*

Conclusion *[A concise, scoped conclusion. Separate issue assessment, verification of a proposed repair, and whether authors have adopted any change.]*

4 AI assistance and contributor declaration

> AI-assisted scientific review is encouraged. Use AI to organize the report, check a derivation or inspect code, then verify the source, assumptions and outputs yourself. Do not treat model agreement as independent evidence. Manual work is equally welcome.

AI used *[Yes or no. If no, delete the AI-specific entries below.]*

Model and access *[Actual provider, model/version, interface or tool, reasoning setting, and date(s) used. Do not invent unavailable settings.]*

Assistance *[Which tasks AI performed; what text, sources or code it saw; material prompts or a concise reproducible description.]*

Human checks *[Which source passages, calculations, citations and outputs you personally verified. Identify unverified AI suggestions explicitly.]*

Conflicts and role *[Authorship, collaboration, financial or other relevant relationships; your role in this analysis.]*

Responsibility *[I checked the cited evidence and accurately described the scope, uncertainty, AI assistance and limitations of this contribution.]*

Attribution *[Contributor name or preferred credit, affiliation if relevant, and date. Do not put private contact details in the report.]*

> List only sources actually consulted. Give authors, title, year, DOI or stable URL and exact location when relevant. Brief quotations must be clearly marked. Use this self-contained bibliography; no separate .bib file is needed.

References

[1] *[Authors. Title. Venue, year. DOI or stable URL and version.]*

[2] *[Additional source actually used. Delete if not applicable.]*